

Strong and Weak Entities in ER Modeling

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Strong Entity

- **What is a Strong Entity?**
- A **strong entity** is an entity that can be uniquely identified by its own primary key and does not depend on any other entity for its existence.
- **Characteristics**
- Has its own **Primary Key**.
- Exists independently.
- Can participate in relationships with other entities.
- Represented by a **single rectangle** in an ER diagram.
- May have one or more attributes.
- Can exist even if related entities are deleted.
- Has either partial or total participation in relationships.

Weak Entity

- **What is a Weak Entity?**
- A **weak entity** is an entity that cannot be uniquely identified using its own attributes. It depends on a strong entity for identification and existence.
- **Characteristics**
- Does not have a complete Primary Key.
- Depends on a strong (owner) entity.
- Has a **Partial Key (Discriminator)**.
- Requires an **Identifying Relationship**.
- Always has **Total Participation**.
- Represented by a **double rectangle**.
- The identifying relationship is represented by a **double diamond**.

Comparison and Real-World Example

Feature	Strong Entity	Weak Entity
Primary Key	Has its own	Uses Owner Key + Partial Key
Existence	Independent	Dependent
Identification	Self-identified	Identified through owner
ER Symbol	Single Rectangle	Double Rectangle
Deletion	Exists independently	Deleted if owner is deleted

Applications

- **Applications of Strong and Weak Entities**
- Hospital Management System
- Banking System
- Library Management System
- University Database
- E-Commerce System
- Railway Reservation System
- Employee Payroll System
- Hotel Management System

Advantages

- **Strong Entities**
- Easy to identify using a unique primary key.
- Can exist independently.
- Simplifies data retrieval and management.
- **Weak Entities**
- Prevent duplicate data.
- Maintain referential integrity.
- Improve database normalization.
- Model dependent objects accurately.
- Reduce data redundancy.